TAIKO · PROTOCOL DESIGN · AUGUST 2026

Based Preconfirmation Redesign

A perpetual auction with commit → publish → seal epochs
Design v9 — a learning deck

Anyone can win the sequencing seat · Slashing from day one · Finality at the seal · No oracles

MOTIVATION

Why redesign preconfirmations?

Where we are today

- Preconfirmations run on a **trusted whitelist** of operators — users trust reputation, not collateral.
- The planned permissionless path required preconfers to be **L1 validators** registered in a Universal Registry Contract (URC) — blocked twice over: little validator adoption, heavy integration burden.
- Slashing machinery exists on paper but is **dormant**: bonds are configured to zero, faults have no teeth.
- A preconf promise today is only as good as the operator's brand.

The redesign's premise

- Decouple the seat from L1 validators: sell sequencing rights in an **open, perpetual on-chain auction**.
- Make every promise **bonded and objectively slashable** from the first day.
- Make finalization **atomic with proving** — the proposal itself carries a validity proof, so there is no separate prover market and no finality lag.
- Design the **failure modes first**: every degraded state has a permissionless exit.

MOTIVATION

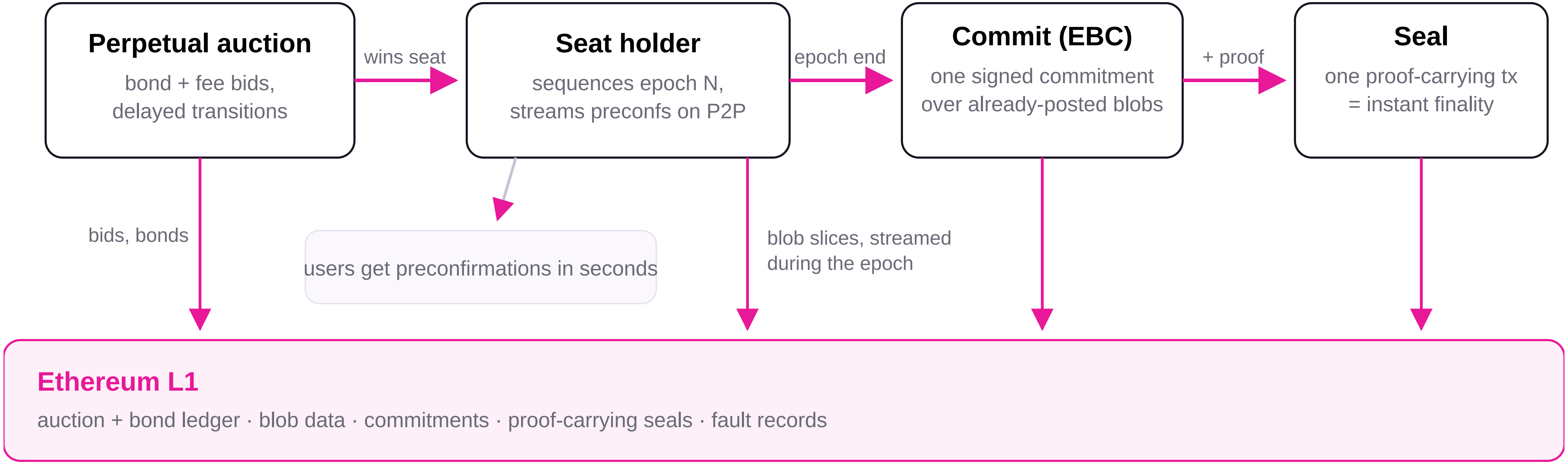
Five goals steer every mechanism

GOAL	WHAT IT DEMANDS
G1	**Anyone can become the preconfer** — a perpetual auction for a single sequencing seat, open to any bonded bidder (no validator requirement).
G2	**Slashing from day one** — on objective faults only: liability is computed from L1 records, never from someone's report.
G3	**Non-validators can act reliably on L1** — every mandatory action has a multi-slot budget; the seal has a multi-epoch window.
G4	**Sealed epochs finalize immediately** — proof verified at acceptance; fast withdrawals; no separate prover market.
G5	**Explicit degradation endgames** — the chain keeps moving through every failure; in the permissionless phase anyone can end a degradation by outbidding the reserve floor.

G5 is a *liveness* guarantee, not a user-fund-safety guarantee: an L1 reorg deeper than the grace κ can revert L2 state exactly as on any rollup.

MENTAL MODEL

The big picture



One seat at a time; one open epoch at a time; every promise backed by bond; every outcome provable by anyone from L1 data.

CAST

Roles and tenures

- **Seat holder** — the current auction winner; sequences epochs, issues preconfs, commits and seals them.
- **Standby bidders** — bonded runners-up; auto-promoted (bindingly) if the holder quits, idles, or is slashed out.
- **Observers** — anyone; earn bounties for recording faults and recovering stalled epochs.
- **Users** — receive preconfirmations; can always force-include through L1.
- **Treasury / DAO** — receives auction fees; governs parameters prospectively.
- **L2 nodes** — re-derive everything from L1; hold nobody's word for anything.

Tenure = the accountable identity

A tenure is an immutable id binding the holder, its registered keys, its assigned epochs, and its reserved collateral. Every artifact it signs binds `(chain, tenure, epoch, role)`.

Every fault gets a stable logical id — `H(chain, tenure, epoch, class, position)` — debited **at most once**, ever.

Two-phase rollout

Phase A: auction entry temporarily allowlisted (training wheels; DAO commits to a fast-path SLA). **Phase B:** fully permissionless. Same mechanics in both.

MECHANISM 1 / 4

The perpetual auction (on L1)

- One sequencing seat, auctioned **perpetually** — the incumbent keeps it until outbid, quitting, or terminated.
- A bid = **TAIKO liveness bond** + **ETH deposit** + **per-epoch ETH fee** paid to the treasury; a new bid must beat the incumbent by **≥ 10%**. The ETH lives in **one account with a seniority waterfall**: recovery tranche → safety tranche → working tranche, under a single admission solvency check.
- Every transition is delayed by `q = 2` epochs — the current and next epochs are always final, so handovers are never a surprise.
- Exits are never free: quitting gives q epochs of notice; **idling pays exactly what quitting would** (the fee clock keeps running), so going dark is never the cheaper exit.
- Termination on: a settled fault, bond below reservation, or `K_empty = 16` idle epochs — and every tenure **expires after at most T_max** (weeks-scale) and re-auctions: the one idleness/censorship bound no Sybil can reset (v9).

Why the auction lives on L1

The auction and the bond ledger must keep operating even when the L2 itself is degraded or halted — so they live on L1. Slashing *evidence* is proven on L2 (cheap, expressive) and the verdict is bridged up to the single L1 ledger.

No perpetual grandfathering

Every *new* epoch assignment — the incumbent's included — is gated on the *current* reservation. Raise the floor, and a holder that does not top up is gracefully rotated out.

MECHANISM 1 / 4 · COLLATERAL

A split bond — and no oracles, anywhere

Safety bond — in ETH

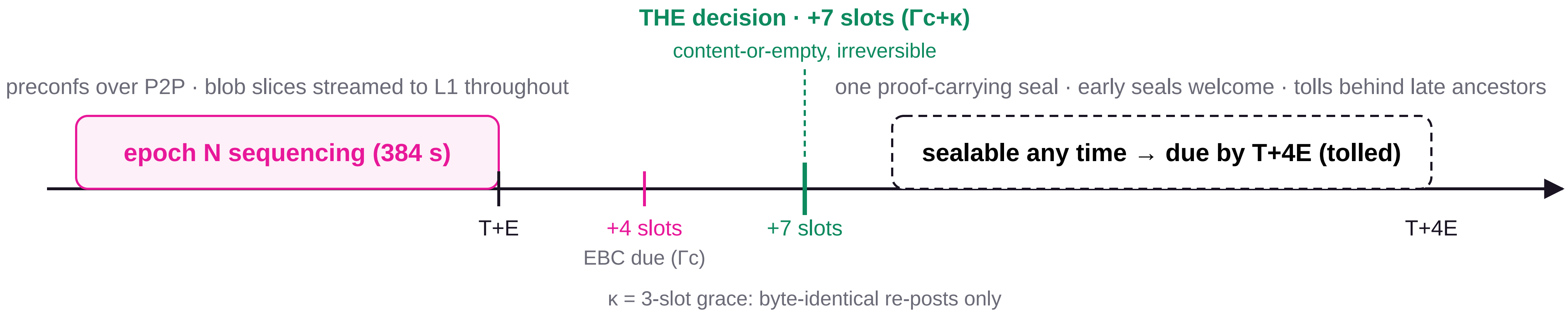
The safety tranche of the tenure's single ETH account. Backs the **theft** class: equivocation / MEV theft, where the attacker's gain is ETH-denominated. Deterrence and gain share the same numéraire, so it is sized to exceed per-epoch extractable MEV **without any price feed** — and its value is fixed from tenure start.

Liveness bond — in TAIKO

Backs the **delay** class: missed commits and seals, which slow the chain but steal nothing. A TAIKO price move weakens griefing deterrence at worst — a bounded, governance-adjustable residual.

- **Owner directives honored together:** TAIKO as the bond token, and no price oracles — ETH is assumed stable-to-increasing.
- A TAIKO price crash cannot cheapen the theft slash; the short-then-misbehave and withdraw-after-shock attacks disappear because the theft debit's value never re-prices.
- Liveness slashes are **≥ 80% burned** — faults must destroy value, not move it.
- All payouts the protocol ever makes go to **precommitted payees** (proof public inputs or pre-registrations) — front-running a copied proof or poke redirects nothing.

One epoch's life on the clock



While epoch N settles, epoch N+1 is already sequencing: its first ~7 slots are **soft** preconfs (parent-provisional), upgrading to **hard** (bond-backed) the moment N's outcome is final. Proving latency (~10–15 min zkVM) fits well inside the 4-epoch seal deadline.

The EBC — one commitment decides the epoch

- The **epoch-boundary commitment** is one-shot per (tenure, epoch) — and **presence at the single decision (+7 slots) is all that counts**: a first submission any time in those 7 slots is valid, so the honest budget is the full span, not just the 4-slot nominal deadline.
- It binds the full ordered content, the end-of-preconf tip, and the **hashes of blob slices already posted to L1** — data first, commitment second.
- It also commits **its own L1 origin** (anchor block hash/state root). Derivation reads the committed origin, never the block that happened to include the transaction — so an L1 reorg can shift *when* things land, never *what* they mean.
- The committed anchor must be ≥ 32 slots deep (reorg safety), fresh, and advancing (no serving from stale L1 state).
- Miss the EBC ⇒ the epoch resolves **EMPTY** and a liveness certificate settles. An explicit empty EBC is valid — unless forced inclusions are pending.

Totality: garbage cannot wedge the chain

Derivation is **total and bounded**: any committed bytes map to exactly one L2 block sequence. Malformed or oversized content degrades *deterministically* to default content — under consensus-exact parse caps (size, RLP count/depth, tx count) enforced identically by client and proof circuit.

Consequence: a committed epoch has **one canonical outcome, provable by anyone holding the data**, no matter when or where anything landed on L1.

Publication is part of the commitment

- There is **no separate publication deadline**: the holder streams its blob data to L1 *during its own epoch*, and the EBC is valid **only if every referenced byte is on L1 at the single decision** (+7 slots) — first inclusions and re-posts alike (v9).
- So content-or-empty is a **single decision, 7 slots after the epoch** — not a second judgment ~27 slots later.
- "Nothing new" means nothing outside the EBC's committed hashes — a delayed slice still lands validly after the boundary, from **any** data holder (P2P spreads them during the epoch).
- If a third party fills a slice the holder failed to keep available, the filler earns a reward — **funded by the failing holder**, paid to a pre-registered beneficiary. No fault, no reward, no honest party to front-run.

Why this kills the "last look"

A predecessor that published on time has **no residual choice left** — its outcome is locked by its own EBC. The successor's exposure to a mischievous parent shrinks to the fixed 7-slot soft-preconf window (~22% of an epoch), down from ~84% under a late-publication design.

Censorship has to beat everyone

Suppressing an epoch means censoring blob posts across a 32-slot span *plus* the re-post grace, against **every holder of the data** — the most expensive artifact to censor, priced explicitly in the game-theory analysis.

MECHANISM 2 / 4 · SEAL

The seal: proving is finalizing

- One small, retryable L1 transaction carries the validity proof for the epoch's canonical outcome; **acceptance = finality** (G4).
- Proofs are fully **precomputable**: content is final 7 slots after the epoch, and the proof binds no later L1 state. The seal has a **deadline only** (due by T+4E, tolled) — sealing early is always valid and shrinks everyone's exposure.
- The seal contributes **zero derivation inputs** — sealing late or from a different sender can never change what the epoch means (I7).
- The payout address is a **public input of the proof**: copying a proof from the mempool advances the chain but pays the original prover's payee (I8).

One open epoch

Epochs seal **strictly in order** through a single `openEpoch` pointer. Every later deadline *tolls* while an ancestor is unresolved — a slow predecessor never converts into a successor's fault.

The recovery lane never closes

At every moment some permissionless action can advance `openEpoch`: the content seal, a forced-only seal, an empty seal, or — as an absolute floor — the bounded cancellation path. Unproven material never blocks any of them (I3).

THE USER'S VIEW

What a preconfirmation means now

MOMENT	THE PROMISE YOU HOLD	BACKED BY
Seconds — P2P preconf	A signed commitment binding (tenure, epoch, position, hashes, deadline) — equivocation is a safety fault.	ETH safety bond (MEV-sized)
First ~7 slots of the next epoch	*Soft* preconf: explicitly provisional on the parent's single decision.	Parent's own EBC + bond
7 slots after the epoch	Content-or-empty is **irreversible**; hard preconfs from here.	Protocol invariant (single decision)
At the seal	Proven, finalized L2 state on L1.	Validity proof

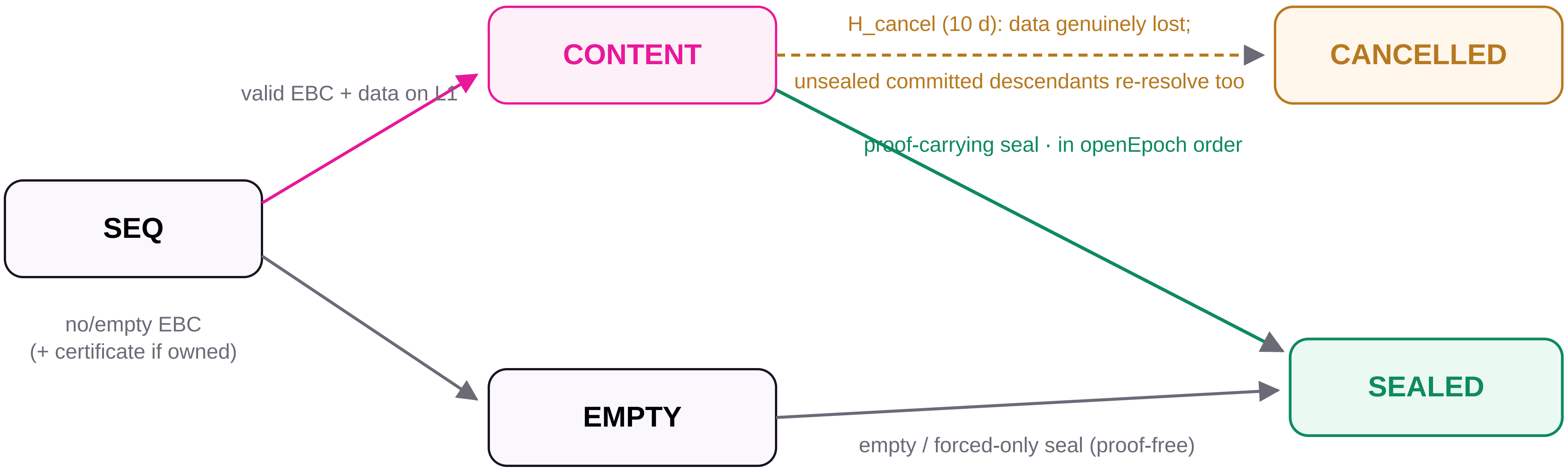
Force-included transactions ride the same rails: once snapshotted, an epoch's minimum valid outcome *includes them* — an empty resolution is simply invalid while the forced queue is non-empty (I6).

THE CONSTITUTION

Nine invariants everything must obey

- **I1 — Total, bounded, content-addressed derivation.** Committed bytes have one canonical outcome, independent of all L1 timing.
- **I2 — No fault requires the accused.** Liability is computed state, matured by deadline + absence; every reader materializes it. Pokes accelerate, never gate.
- **I3 — Single open epoch, always advanceable.** A permissionless advancing action always exists.
- **I4 — Successor-safe parent.** Outcomes irreversible at $\Gamma c+\kappa$; deadlines toll behind blocked ancestors.
- **I5 — Bounded global backlog.** Lag $> K \Rightarrow$ recovery-only mode; the unsealed content tail is structurally capped.
- **I6 — Forced inclusions are censorship-proof.** Non-empty forced queue $\Rightarrow$ empty outcomes invalid; the forced-only epoch is constructible by anyone.
- **I7 — Only proven transitions lock state; outcomes are sender-free.**
- **I8 — Payees are precommitted, not first-claimers.** Every reward is bound before the witness is public.
- **I9 — Deterrence value is oracle-free.** ETH-matched slashes are ETH-denominated; no mechanism reads a price.
- **Corollary — sealed epochs are final.** No mechanism, including disaster cancellation, ever touches sealed state.

The epoch state machine



The decision (7 slots) picks CONTENT or EMPTY **once**; seals close epochs **strictly in openEpoch order**; sealed state is untouchable, forever.

The disaster path is the only exception route: cancelling a stuck epoch marks every unsealed descendant commitment that chained to it VOID (each then closes as cancelled, in order), re-queues their forced items at the front, and bills the tenure that caused it — sealed history is never touched.

Forced inclusion: the floor nobody can lower

- Users enqueue transactions on L1 with a fee `a + b·bytes + c·declared_gas` — L1-measurable quantities with conservative constants that upper-bound proving cost; the per-item base makes flooding pay per item.
- Each epoch takes a deterministic **snapshot** of the queue head; overflow spills to the next epoch.
- While the snapshot is non-empty, the epoch's minimum valid outcome is the **forced-only epoch** — constructible and provable by *anyone* from L1 data alone. Empty is invalid (I6).
- Fees flow to whoever seals the consuming epoch — the fallback funds its own proving.

One lifecycle, no double-spends

Every forced item advances `queued → snapshotted → consumed | refunded` on L1, with its payer stored at enqueue time. Seal proofs take the item's status as a public input; a refund atomically kills every live commitment containing the item. Refund *and* execution can never both happen.

Honest accounting (v9)

No chain can detect Sybil "demand", so `K_empty` is only a nuisance bound against lazy idlers — the binding bound is `T_max` tenure expiry. Forced-only fee income per tenure is capped, and a holder sealing its own epochs earns no recovery bounty. Squatting has no profit path and a hard clock.

Faults are computed, never reported

- A liveness fault is an **objective L1 fact**: assignment + deadline + absence. It **matures** the instant its deadline-plus-grace passes — no transaction needed, no cooperation from the accused.
- Anyone may record ("poke") a fault for a bounty, but pokes only *accelerate*: every function that depends on fault status — withdrawal, sealing an empty outcome, promotion — **computes maturity itself and materializes the certificate as part of the call**.
- Consequence: censoring pokes buys nothing. A holder that missed an obligation *faults itself by trying to withdraw*.
- Each deadline has exactly **one κ grace** and **one atomic transition**: present ⇒ accepted; absent ⇒ certificate settled, parent outcome fixed, slashability locked — all at the same instant.

Content faults: evidence on L2, money on L1

Equivocation and content faults are proven cheaply on L2 by permissionless observers; the verdict is bridged to the **single L1 bond ledger**, which alone enforces one-debit-per-fault. Verdicts bind a finalized origin root + incarnation number, so a verdict minted on an orphaned fork is dead on arrival after a deep reorg.

Withdrawal gate

Exit requires: zero unresolved certificates (computed, not poked), no unsealed assigned epochs, no in-flight verdicts, then a ≥ 2-week floor. Bonds outlive every obligation they back.

When a holder fails, the chain barely notices

The failure playbook

- **Missed commit** → epoch resolves EMPTY at the single decision; certificate settles; slash debits; chain continues.
- **Missed seal** → the epoch's data is on L1, so from the moment the deadline fault matures **anyone** seals it — paid at indexed cost *from the faulter's own recovery tranche*, never from honest tenures.
- **Termination** → the highest standby is auto-promoted (bindingly, after the q-delay); no re-auction gap. No standby ⇒ Total Anarchy (next slides).
- **Every recovery is compensated, never framed**: certificates name only the tenure that actually held the duty; deadlines toll behind blocked ancestors.

Solvent by admission

The senior tranche of each tenure's single ETH account is sized to its worst-case K outstanding recovery obligations, checked at admission: `Σ reserves ≥ Σ worst-case obligations`. A deliberate fault can only ever spend its own account; a thin shared pool covers only the honest remainder.

Recovery pays cost, not prizes

Compensation is a capped multiple of prevailing L1 + proving costs (indexed, not fixed) — gas spikes raise it, and there is no margin worth attacking for. Deterrence lives in the ≥ 80% burn, not the bounty.

DEGRADATION

A ladder, not a kill switch

Rung 1 · The seal deadline — fault-paid resolution, nothing global

The moment a stuck epoch's seal deadline passes, its fault matures and anyone seals it from L1 data, paid from the faulter's recovery tranche. One bad holder is never anyone else's problem — and no separate horizon is needed.

Rung 2 · Recovery-only mode — when lag exceeds K = 8 epochs

New discretionary content pauses; all effort funnels into the recovery lane. No withdrawal freeze — honest tenures with clean books may still exit.

Rung 3 · Attested systemic-outage freeze — never reachable by one holder

A global pause of maturation + withdrawals requires an independent attestation of a genuine proof-system outage — never openEpoch age alone. Forgiveness never covers whoever drove the stall.

Absolute floor: H_cancel (10 days, data genuinely unavailable) — the epoch re-resolves forced-only/empty, unsealed descendants re-resolve deterministically, and the causing tenure pays for the whole cascade. Sealed history is never touched.

Total Anarchy — the fallback with an exit sign

- No holder and no standby ⇒ epochs are unowned: they resolve empty or **forced-only**, sealed permissionlessly — the bridge and forced lane keep flowing at L1 cadence.
- No discretionary content, no preconfs, no slashing — nothing is bonded to promise anything. (A v7 attempt to restore FCFS content via atomic proposals was **reverted by the v8 regression audit**: a cheap empty seal beats any ~15-minute proof, re-opening a censorship lever an earlier review round had closed.)
- **Phase B exit:** anyone ends anarchy by bidding the reserve floor — a censorship-resistant fallback, not a sequencing mode.
- **Phase A exit:** the DAO's pre-committed fast-path SLA expands the allowlist within a published deadline.

Deposits stay safe, and bounded

Even a worst-case voided forced item settles as a refund: queue fees refundable on L1, bridged messages refundable through the bridge's unprocessed-message path. Worst-case bridge settlement is **H_cancel + one forced-only cadence** — a number a depositor can price, not an open-ended limbo. Value is only ever delayed, never burned.

Why anarchy is acceptable

It is the *floor*, not the plan: reaching it requires the seat, every standby, and every would-be bidder to decline a profitable seat — and staying in it is exactly as hard as out-bidding a floor price.

ADVERSARIAL ECONOMICS

Attacks the design prices out

ATTACK	WHY IT FAILS
Preconf equivocation / MEV theft	ETH safety bond sized above per-epoch extractable MEV; value fixed from tenure start — no oracle, no shorting window, no price-crash discount.
Squat the seat, serve nobody	Idling pays quit-equivalent fees; `K_empty` nuisance-terminates; `T_max` expiry evicts even a perfect Sybil (v9); forced-only income capped.
Withhold the seal, hold users hostage	Data is already on L1; the deadline's matured fault lets anyone seal at the faulter's expense. A stall buys ~minutes, and costs a slash.
Censor the poke, then exit	Liability is computed at read time — withdrawal itself materializes the fault (I2).
Front-run proofs / pokes / fills	Payees are precommitted (proof public inputs, pre-registrations) — copying a witness pays the original beneficiary (I8).
Flood forced inclusions	Per-item base fee + byte/gas-proportional pricing bounds proving cost; snapshots cap per-epoch load, overflow spills deterministically.
Publish a long tail, then cancel it	Recovery-only mode caps the unsealed content tail at ≈ K+d+s epochs (~77 min) — a 14-day cancellation avalanche cannot exist.

Stated residuals: sub-H_force targeted censorship can cost an honest holder one liveness slash (priced unprofitable); TAIKO-price erosion weakens only griefing deterrence; K_empty bounds self-serving, but a complete censorship bound awaits the max-tenure-duration decision.

CONFIDENCE

How hard has this been attacked already?

11 passes

of adversarial review across 7 rounds (multiple independent AI reviewers), every finding accepted-or-rebutted in a public disposition table. An owner-approved simplification round (v7) was **re-audited against every prior finding** — the one change that regressed was reverted (v8) — and round 7's liveness blockers landed T_max, decision-time availability, and the outage-robust checker (v9).

6.0 M states

explored by an exhaustive model checker of the epoch state machine — **zero invariant violations, zero permanent halts, even under a proving outage that never lifts** (v9): the proof-free floor of empty seals and cancellation always finalizes the chain.

The checker checks itself

9 deliberately-injected design bugs (double decision, ignore-forced, out-of-order seal, ungated withdrawal, missing cascade, double debit, evidence erasure, unmaterialized fault, outage-blocks-everything) — **9/9 caught** by the check written to catch each.

BOUND (V9, OUTAGE ADVERSARY)	STATES	VIOLATIONS	HALTS
3 × 4	2,339,418	0	0 / 0
3 × 5	3,659,594	0	0 / 0

Halts column = standard / outage-robust passes. Bounded checking of a logical abstraction — strong evidence for the state machine, not a proof of the eventual Solidity.

WHERE IT STANDS

Status and reading map

Before implementation (§13-S)

- Deep-reorg (> κ) joint-rewind semantics; cascade determinism proof.
- Consensus-exact parse caps across client and circuit; derivation-input sign-off (Appendix C).
- Record-spine retention/compaction; recovery-tranche sizing (incl. the cascade charge); forced-item nullifier; verdict incarnation; precommitted-payee binding; the T_max re-auction mechanism.

Tuning before Phase B (§13-T)

- Soft-window width vs duty-cycle; max tenure duration (the complete censorship bound); K/K'/K_empty/H_cancel calibration; Phase A→B criteria.

Read the full story

`docs/preconfirmation/README.md` — index

`status-quo.md` — what exists today, premise validation

`redesign-proposal.md` — the v6 design (this deck's source of truth)

`simulation/` — model checker + results

`reference/` — the two prior Notion design docs, converted

Discuss

PR **taikoxyz/taiko-mono #22034** — all review rounds, defenses, and dispositions are on the thread and in Appendix B.